

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 1

Design Desk

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines, and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Assessment Tool Weightage: 30%

QUESTIONS:

1. Design an Autoencoder-Based Feature Learning System

A manufacturing company wants to detect defective products from high-dimensional sensor data. Design an Undercomplete Autoencoder with suitable encoder, latent representation, decoder, activation functions, and reconstruction loss. Explain how the design performs dimensionality reduction and feature learning.

2. Design a Robust Regularized Autoencoder

A healthcare organization has noisy patient-record features and needs a robust representation for downstream classification. Design a Regularized Autoencoder using an appropriate regularization strategy. Explain the encoder-decoder structure, regularization mechanism, training objective, and how the design reduces overfitting and improves useful feature extraction.

3. Design a Deep Generative Model

A recommendation or content-generation system requires learning complex data distributions. Design a Deep Belief Network or Deep Boltzmann Machine using suitable visible and hidden layers. Illustrate the architecture, explain the learning process, and justify the choice of the model for the proposed application.

4. Design a Stochastic and Contractive Representation Model

A real-world image or signal-processing application requires representations that remain useful under small input variations. Design a model using Stochastic Encoders/Decoders or a Contractive Encoder. Specify the architecture, objective function, and robustness mechanism, and explain how the design improves representation stability.

5. Design a GAN-Based Data Generation System

An organization has limited training images and wants to generate realistic synthetic samples. Design a Generative Adversarial Network with a Generator and Discriminator. Specify the input, network flow, adversarial training process, loss functions, and evaluation approach. Compare the proposed GAN design with an Autoencoder and justify why GAN is suitable for the application.

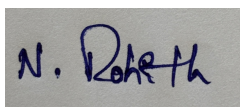
Design Desk Guidelines

- Identify the real-world problem, requirements, inputs, outputs and constraints.
- Draw a clear architecture or workflow diagram for the proposed model.
- Specify important layers, units, activation functions, latent representation and loss functions.
- Explain the training or learning mechanism and justify major design choices.
- Mention suitable evaluation measures and expected outcomes.
- Use neat labeling and present the design in a logical sequence.
- The solution should be original and written in the student's own words.

Code of Conduct:

I **NALIPI REDDY ROHITH/192425115** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem, requirements, constraints, and expected outcomes.	Good understanding with minor gaps in requirements.	Basic understanding; some requirements are unclear.	Poor understanding or misinterpretation of the problem.
Application of Concepts	30	Accurately applies appropriate generative deep learning concepts and architectures.	Applies relevant concepts correctly with minor errors.	Applies basic concepts with limited accuracy.	Fails to apply appropriate concepts.
Design Approach	20	Innovative, logical, well-structured, feasible, and technically justified design.	Logical and feasible design with minor gaps.	Basic design with some inconsistencies.	Unclear, illogical, or impractical design.
Accuracy of Solution	20	Completely correct architecture, process, objectives, and justification.	Mostly correct with minor technical errors.	Partially correct solution with noticeable gaps.	Incorrect or incomplete solution.
Clarity	10	Clear, well-organized diagram and explanation with proper technical labeling.	Understandable with minor clarity issues.	Limited clarity and explanation.	Poorly presented and difficult to understand.

Design Desk

Q1. Design an Autoencoder-Based Feature Learning System

1. Problem

A manufacturing company has high-dimensional sensor data such as temperature, pressure, vibration and speed. The objective is to reduce the number of features and learn useful representations for detecting defective products.

2. Proposed Architecture

Input Sensor Data



Dense Layer – 128



Dense Layer – 64



Latent Representation – 16



Dense Layer – 64



Dense Layer – 128



Reconstructed Sensor Data

3. Design

- **Input:** High-dimensional sensor features
- **Encoder:** $128 \rightarrow 64 \rightarrow 16$ neurons

- **Latent representation:** 16-dimensional vector
- **Decoder:** $16 \rightarrow 64 \rightarrow 128 \rightarrow$ original input size
- **Hidden activation:** ReLU
- **Output activation:** Linear
- **Loss:** Mean Squared Error (MSE)

4. Working

The encoder converts the original high-dimensional input into a smaller latent representation. This latent vector stores the most important information from the sensor data.

The decoder reconstructs the original input from this compressed representation.

The model minimizes:

$$L = \frac{1}{n} \sum (x - \hat{x})^2$$

where (x) is the original sensor data and $(\hat{x})$ is the reconstructed data.

5. Dimensionality Reduction

If the original sensor data contains 100 features, the encoder can compress them into only 16 latent features.

$100 \text{ Features} \rightarrow 64 \rightarrow 32 \rightarrow 16 \text{ Features}$

Thus, irrelevant and redundant information is reduced while important patterns are retained.

6. Evaluation

Suitable measures are:

- Reconstruction MSE
- MAE
- Classification accuracy after using latent features
- Precision and recall for defect detection

Expected outcome: The system produces compact features that can be used for efficient defect detection.

Q2. Design a Robust Regularized Autoencoder

1. Problem

Healthcare records may contain noisy and irrelevant information. A regularized autoencoder can learn useful patient representations while reducing overfitting.

2. Architecture

Noisy Patient Data



Dense 128 + ReLU



Dense 64 + ReLU



Latent Layer 16



Dense 64 + ReLU



Dense 128 + ReLU



Reconstructed Data

3. Regularization Strategy

A suitable choice is **Dropout and L2 regularization**.

- Dropout randomly removes some neurons during training.
- L2 regularization penalizes very large weights.

The objective becomes:

$$L = \text{MSE}(x, \hat{x}) + \lambda \sum w^2$$

4. Training

The noisy patient records are provided to the encoder. The encoder produces a compact latent representation. The decoder reconstructs the original clean representation.

The model learns to retain useful information while ignoring noise.

5. Advantages

- Reduces overfitting.
- Improves robustness to noisy records.
- Produces compact features.
- Useful for downstream disease classification.
- Prevents the network from simply memorizing training data.

6. Evaluation

Use:

- Reconstruction MSE
- MAE
- Classification accuracy
- Precision
- Recall
- F1-score

Expected outcome: The regularized autoencoder produces stable and useful patient representations for classification.

Q3. Design a Deep Generative Model

1. Problem

A recommendation system needs to learn complex patterns in user preferences and generate suitable recommendations.

A **Deep Belief Network (DBN)** can be used to learn hierarchical representations from user-item interaction data.

2. Architecture

User-Item Input



Visible Layer



Hidden Layer 1



Hidden Layer 2



Hidden Layer 3



Recommendation Output

For example:

500 Input Features



256 Hidden Neurons



128 Hidden Neurons



64 Hidden Neurons



Recommendation

3. Model Components

A DBN consists of multiple layers of **Restricted Boltzmann Machines (RBMs)**.

Each RBM contains:

- Visible units
- Hidden units

There are no connections between units within the same layer.

4. Learning Process

Training occurs in two stages:

Stage 1 – Unsupervised Pre-training

Each RBM learns representations from the data. The hidden representation of one RBM becomes the input to the next RBM.

Stage 2 – Fine-tuning

The complete network is fine-tuned for the recommendation task.

5. Why DBN?

DBNs can learn hierarchical patterns from complex data without requiring all training data to be labeled.

For a recommendation system, the model can learn hidden relationships between users and items.

6. Evaluation

Use:

- Recommendation accuracy
- Precision@K
- Recall@K
- F1-score
- RMSE for rating prediction

Expected outcome: The DBN learns hidden user-preference patterns and produces personalized recommendations.

Q4. Design a Stochastic and Contractive Representation Model

1. Problem

In image and signal processing, small variations such as noise, brightness changes or sensor fluctuations should not completely change the learned representation.

A **Contractive Autoencoder (CAE)** can be designed to produce stable representations.

2. Architecture

Input Image / Signal



Dense/Conv Layer



ReLU



Latent Representation



Decoder

↓

Reconstructed Input

Example:

784 Input Pixels

↓

256 ReLU

↓

64 ReLU

↓

16 Latent Features

↓

64 ReLU

↓

256 ReLU

↓

784 Output

3. Objective Function

A contractive autoencoder combines reconstruction loss with a penalty on the encoder's sensitivity.

$$\begin{aligned} [\\ L = L_{\text{reconstruction}} + \\ \lambda \|J\|_F^2 \\] \end{aligned}$$

where (J) is the Jacobian of the encoder.

4. Robustness Mechanism

The contractive penalty forces the representation to change only slightly when the input changes slightly.

For example:

Original Image $\rightarrow$ Representation A

$\downarrow$

Slightly Noisy Image $\rightarrow$ Representation A'

A good CAE makes A and A' very similar.

5. Advantages

- Robust to small input variations.
- Reduces sensitivity to noise.
- Learns stable features.
- Useful for image and signal processing.
- Improves representation quality.

6. Evaluation

Use:

- Reconstruction error
- Classification accuracy
- Noise robustness
- Feature stability

Expected outcome: The model learns representations that remain stable even when small noise or variations are introduced.

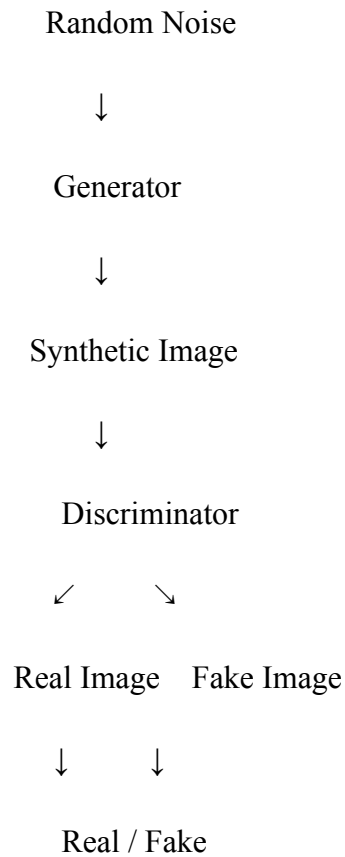
Q5. Design a GAN-Based Data Generation System

1. Problem

An organization has only a small number of training images. A **Generative Adversarial Network (GAN)** can generate additional realistic synthetic images.

2. Architecture

A GAN consists of two neural networks:



3. Generator

The Generator receives a random noise vector:

$$\begin{bmatrix} z \end{bmatrix} \rightarrow G(z)$$

and converts it into a synthetic image.

Example:

Noise Vector



Dense Layer



Upsampling



Convolution



Synthetic Image

The generator attempts to produce images that look real.

4. Discriminator

The discriminator receives both real and generated images.

Image



Convolution



Feature Extraction



Dense Layer



Real / Fake

Its objective is to correctly distinguish real images from generated images.

5. Adversarial Training

The two networks compete with each other.

- Generator tries to **fool** the discriminator.
- Discriminator tries to **detect fake images**.

The standard GAN objective is:

$$\begin{aligned} & [\\ & \min_G \max_D \\ & V(D,G) = E[\log D(x)] \\ & + E[\log(1 - D(G(z)))] \\ &] \end{aligned}$$

Through repeated training, the Generator gradually produces more realistic images.

6. Evaluation

Generated images can be evaluated using:

- Visual quality
- Inception Score (IS)
- Fréchet Inception Distance (FID)
- Diversity of generated samples
- Performance improvement of a classifier trained with synthetic data

7. GAN vs Autoencoder

Feature	Autoencoder	GAN
Main purpose	Reconstruction	Data generation
Encoder	Yes	No
Decoder/Generator	Decoder	Generator
Discriminator	No	Yes

Image quality	Usually moderate	Often more realistic
Training	Relatively stable	More difficult
Synthetic data	Possible	Main purpose

8. Why GAN?

GANs are suitable when the primary requirement is to generate **realistic and diverse synthetic images**. The adversarial training process encourages the generated samples to resemble the real training distribution.

9. Practical Workflow

Limited Real Images



Train GAN



Generate Synthetic Images



Combine Real + Synthetic Data



Train Classification Model



Improved Performance

10. Conclusion

A GAN can increase the effective size of a limited image dataset by generating realistic synthetic samples. With proper evaluation and quality control, these samples can improve the training of downstream deep-learning models.